

Statistical Post-Processing of Seoul's LDAPS Weather Forecast

Regression-Based Bias Correction of Numerical Weather Prediction Model Temperature Forecasts

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Abstract

South Korea's Local Data Assimilation and Prediction System (LDAPS) has a persistent blind spot: it consistently underestimates the next day's maximum temperature across all 25 of Seoul's automatic weather stations. This project sets out to understand exactly how large that gap is — and to close it. The difference between actual temperature and predicted temperature by LDAPS (termed the "bias") averages +0.621°C with a standard deviation of 1.734°C- small enough to seem harmless, but meaningful in practice. To correct for it, a statistical post-processing framework was built using a set of predicted: LDAPS's other atmospheric forecasts, present day's situations, and the geographical characteristics of each station. The model was constructed through five deliberate stages — starting from a broad linear baseline, assessing for nonlinear relationships, enriching the model with polynomial and interaction terms, trimming it down using AIC-based stepwise regression, and finally refining it with weighted least squares on mean-centered predictors. The resulting model, wls_final, explains roughly a third of the bias variance (Adj. $R^2 = 0.328$). When this bias correction is applied and the corrected forecasts are evaluated against raw LDAPS output, the improvement is clear. The corrected forecast effectively eliminates systematic error ($E[bias] \approx 0$), lifts predictive accuracy ($R^2 = 0.766$ vs. 0.698), and cuts forecast error from 1.709°C down to 1.504°C RMSE — comfortably surpassing the Cho et al. (2020) benchmark in the process.

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Chapter 1

Introduction

Accurate short-range temperature forecasting is critical for public health, energy-grid management, and agricultural planning. South Korea's Korea Meteorological Administration (KMA) operates the **Local Data Assimilation and Prediction System (LDAPS)**, a high-resolution numerical weather prediction (NWP) model. Despite its high spatial resolution, LDAPS exhibits a systematic **cold bias**: the model consistently under-forecasts observed next-day maximum temperatures, with a published RMSE of $\approx$ 2.08 °C and published $R^2 \approx 0.69$ (1).

Statistical post-processing addresses this problem without modifying NWP physics: a regression model learns and corrects the systematic error (bias) at each station. This project applies multiple regression to accomplish exactly that goal, using the Cho et al. (2020) published benchmark as an explicit target.

1.1 Research Objectives

1. Characterise the structure, magnitude, and predictors of the LDAPS temperature bias through exploratory and inferential analysis.
2. Construct the best-possible linear regression bias model and validate on a test set.
3. Construct the corrected temperature forecast and demonstrate its advantage over the raw LDAPS output via direct regression comparison.

1.2 External Benchmark

The published LDAPS performance (1) provides a verifiable baseline: $R^2 \approx 0.69$, RMSE $\approx$ 2.08 °C. A corrected model achieving $R^2 \geq 0.75$ and RMSE $\leq$ 1.70 °C represents a meaningful and replicable improvement using linear regression alone.

Chapter 2

Data and Pre-Processing

2.1 Dataset Description

The *Bias Correction of Numerical Prediction Model Temperature Forecast* dataset (2; 1) (UCI Repository #514) contains **7,752 summer observations** (June–August) from **2013 to 2017** across **25 automatic weather stations** in Seoul, South Korea. After removing rows with any missing model-variable the analytic dataset comprises $\mathbf{n = 7{,}588}$ complete observations. Table 2.1 summarises the variable groups.

Table 2.1: Variable groups in the LDAPS bias correction dataset.

Group	**Key Variables**	**Description**
	`Next_Tmax`	Observed next-day maximum temperature (°C)
LDAPS forecasts	`LDAPS_Tmax`, `LDAPS_Tmin`	Lapse-rate-corrected temperature forecasts (°C)
LDAPS humidity	`LDAPS_RHmin`, `LDAPS_RHmax`	Min/max relative humidity forecasts (%)
LDAPS atmospheric	`LDAPS_WS`, `LDAPS_LH`	Wind speed (m s^{-1}), latent heat flux (W m^{-2})
LDAPS cloud/precip	`LDAPS_CC1-CC4`, `LDAPS_PPT1-PPT4`	Cloud cover and precipitation in 4×6-hr windows
In-situ persistence	`Present_Tmax`, `Present_Tmin`	Today's observed temperatures (°C)
Geography	`DEM`, `Slope`, `Solar_rad`, `lat`, `lon`	Elevation (m), terrain slope (°), solar radiation, coordinates

2.2 Data Processing

(i) **Cloud cover and precipitation aggregation.** The four six-hourly windows were averaged: `LDAPS_CC_avg` $= \frac{1}{4}\sum_{k=1}^{4} \mathrm{CC}_k$, `LDAPS_PPT_avg` $= \frac{1}{4}\sum_{k=1}^{4} \mathrm{PPT}_k$. The individual windows are highly collinear ($r > 0.90$); aggregation reduces the predictor space without information loss.

(ii) **Elevation classification.** Each station was classified into three DEM terciles: *Low*, *Medium*, *High* using equal quantile break. Low was set as the reference level.

(iii) **Response variable.** $\text{Bias}_i = \text{Next_Tmax}_i - \text{LDAPS_Tmax}_i$. A positive value indicates the LDAPS forecast was too cold.

(iv) **Train/test split.** The dataset was splited by 80/20 (seed = 42) split. Training set: $n = 6{,}075$; test set: $n = 1{,}513$.

Chapter 3

Exploratory Data Analysis

3.1 Bias Distribution

Figure 3.1 shows the distribution of LDAPS forecast bias. The overall descriptive statistics (full dataset, $n = 7{,}588$) are:

	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Bias (°C)	−7.71	−0.52	+0.57	+0.621	+1.76	+9.17

The positive mean (+0.621 °C) confirms the systematic cold bias: on average, LDAPS forecasts are 0.621 °C colder than observed next-day maxima. The distribution is approximately symmetric and unimodal, with mild heavier tails than a Normal distribution.

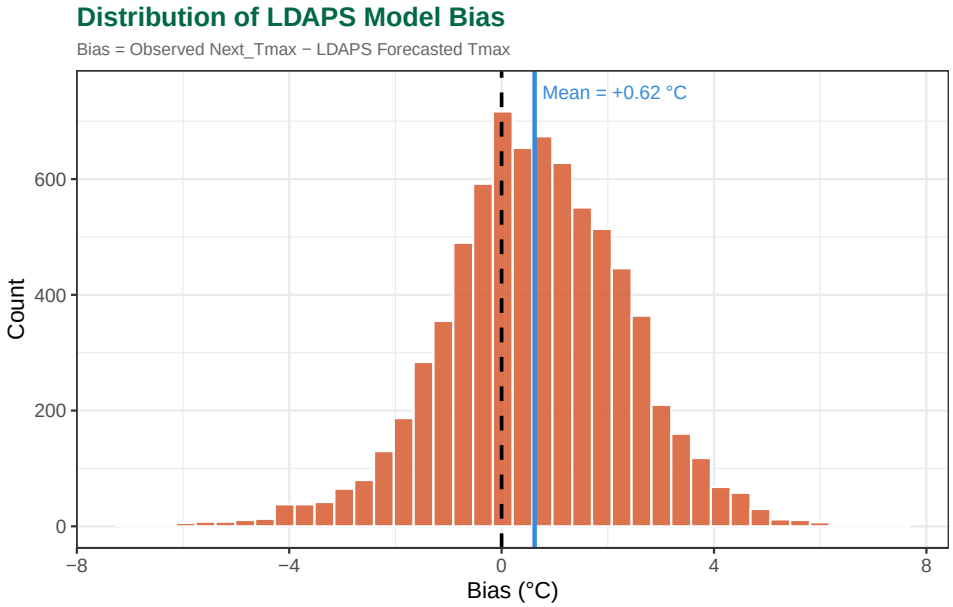


Figure 3.1: Distribution of LDAPS model bias (Bias = Next_Tmax − LDAPS_Tmax).

3.2 Baseline LDAPS Performance

Figure 3.2 shows the LDAPS forecast against observed temperature. The simple regression `Next_Tmax` $\sim$ `LDAPS_Tmax` yields: $\hat{\beta}_0 = 4.072$, $\hat{\beta}_1 = 0.884$, $R^2 = 0.698$, RMSE = 1.709 °C, RSE = 1.709 °C. The OLS fit lies below the 1:1 perfect forecast line throughout the temperature range, visually confirming the cold bias.

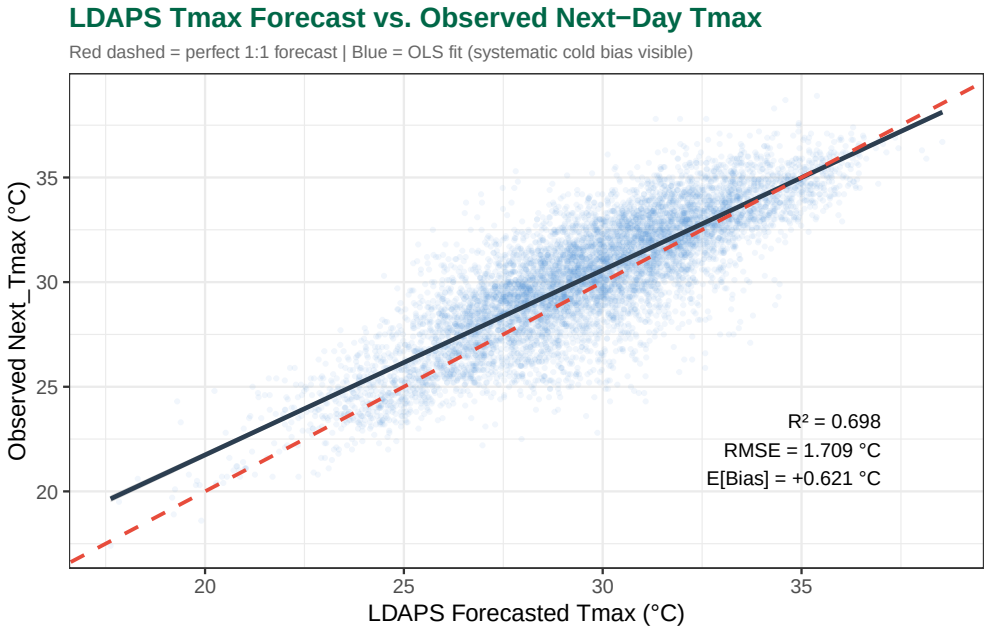


Figure 3.2: LDAPS forecasted Tmax vs. observed Next_Tmax (n = 7,588). Red dashed = perfect 1:1 forecast; solid black = OLS fit. The systematic cold bias is visible: the OLS line consistently falls below the diagonal. Baseline: $R^2 = 0.698$, RMSE = 1.709 °C.

3.3 Bias by Elevation Class and Month

Tables 3.1 and 3.2 give stratified bias statistics by elevation class and month respectively, and Figure 3.3 shows the corresponding boxplots.

Table 3.1: Descriptive statistics of LDAPS temperature bias by elevation class (n = 7,588).

Elevation class	n	**Mean (°C)**	**SD (°C)**	**Median (°C)**
Low	2,625	+0.479	1.669	+0.432
Medium	2,455	+0.573	1.756	+0.527
High	2,508	+0.763	1.772	+0.700

Table 3.2: Descriptive statistics of LDAPS temperature bias by month ($n = 7{,}588$).

Month	n	Mean (°C)	SD (°C)	Median (°C)
June	2,500	+1.052	1.704	+1.019
July	2,546	+0.101	1.748	+0.091
August	2,542	+0.717	1.659	+0.694



Figure 3.3: LDAPS model bias stratified by elevation class (left) and by month (right).

High-elevation stations exhibit a systematically larger cold bias (+0.763 °C) than low-elevation stations (+0.479 °C). Seasonally, June shows the largest positive bias (+1.052 °C), declining sharply in July (+0.101 °C) and recovering slightly in August (+0.717 °C).

3.4 Bias Trend by Year

Figure 3.4 shows the year-by-year mean bias for each elevation class. The inter-annual variability and the persistent elevation gradient are both visible, suggesting that the correction must account for both spatial and temporal atmospheric conditions.

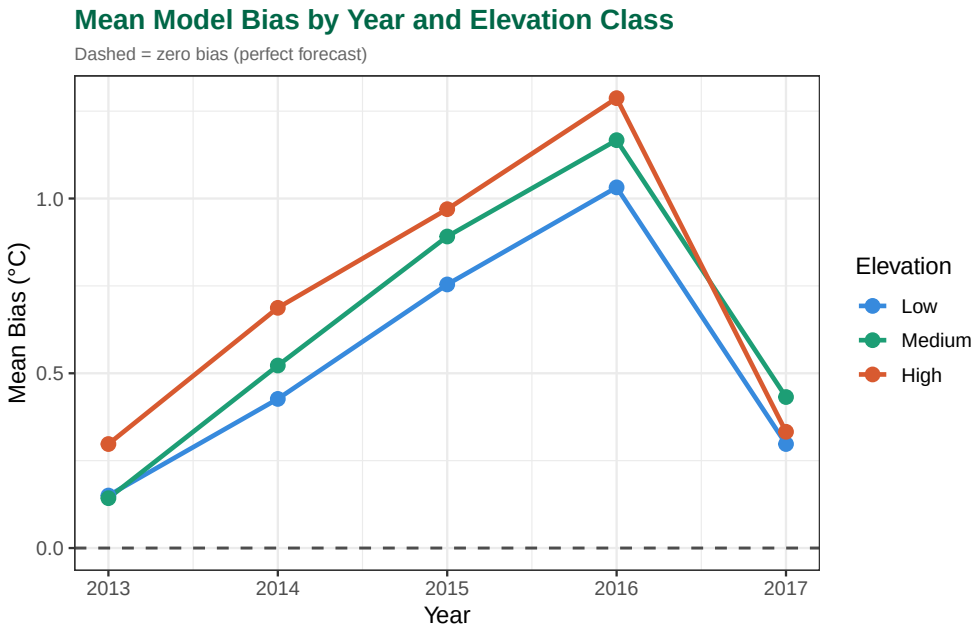


Figure 3.4: Mean LDAPS model bias by year and elevation class (2013–2017). The dashed line marks zero (perfect forecast). High-elevation stations consistently lie above low-elevation stations; all classes show inter-annual variability.

3.5 Predictor Correlation Structure

Figure 3.5 shows the correlation heatmap. `Present_Tmax` and `LDAPS_Tmin` are strongly correlated ($r \approx 0.77$), reflecting the physical autocorrelation of temperature. Bias shows moderate positive correlations with humidity variables (`LDAPS_RHmin`, `LDAPS_RHmax`) and positive correlation with present-day temperature and latent heat.

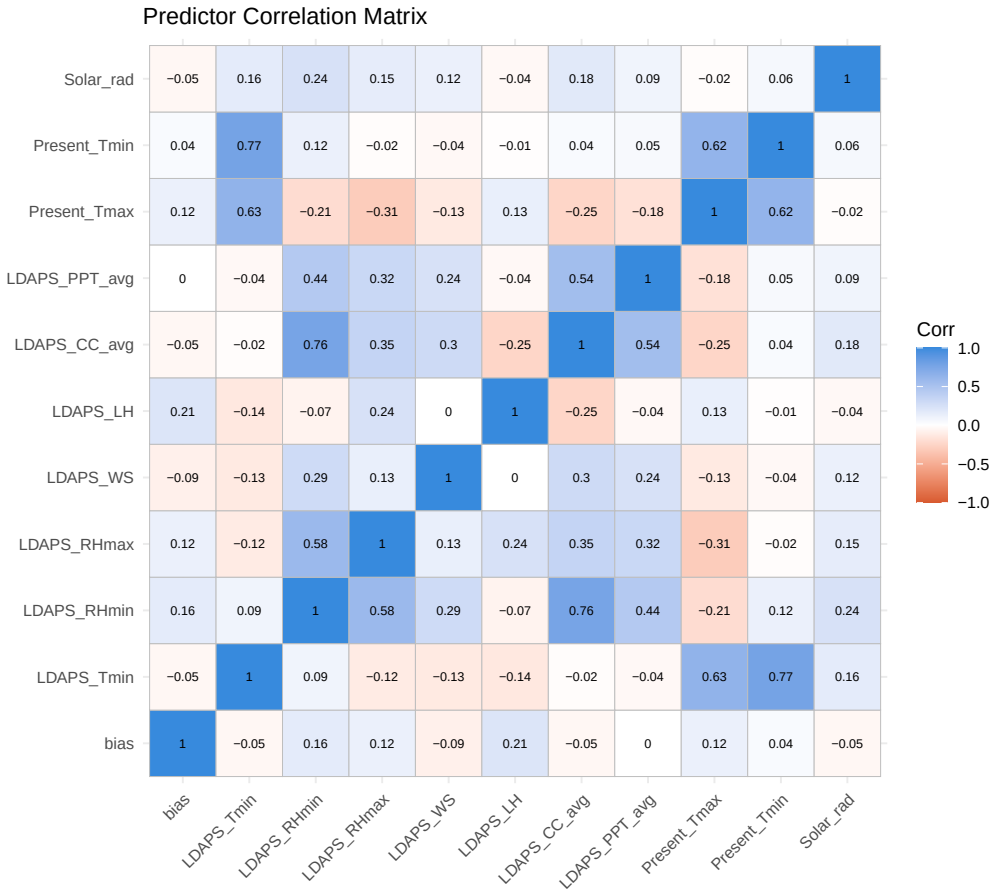


Figure 3.5: Correlation matrix of key numeric predictors and bias.

Chapter 4

Hypothesis Testing: ANOVA by Group

Before fitting regression models, one-way ANOVA confirms that both elevation class and month are statistically significant predictors of mean bias.

4.1 Bias by Elevation Class

Hypotheses:
$H_0 : \mu_{\text{Low}} = \mu_{\text{Medium}} = \mu_{\text{High}}$ vs. H_1: at least one group mean differs.

The one-way ANOVA yields: $F(2,\ 7{,}585) = 11.43$, $p < 0.001$. Tukey's HSD post-hoc test reveals that the High–Low pair is the only pairwise difference reaching significance ($\Delta\mu = +0.284\,^{\circ}\text{C}$, $p = 0.0005$); Medium–Low ($p = 0.104$) and High–Medium ($p = 0.208$) are not individually significant at $\alpha = 0.05$.

4.2 Bias by Month

Hypotheses:
$H_0 : \mu_{\text{June}} = \mu_{\text{July}} = \mu_{\text{August}}$ vs. H_1: at least one group mean differs. The one-way ANOVA with month as factor: $F(2,\ 7{,}585) = 19.28$, $p < 0.001$. Tukey's HSD shows significant differences for July–June ($\Delta = -0.952\,^{\circ}\text{C}$, $p < 0.001$) and August–June ($\Delta = -0.836\,^{\circ}\text{C}$, $p < 0.001$), with a smaller but significant August–July difference ($\Delta = +0.116\,^{\circ}\text{C}$, $p = 0.011$).

Chapter 5

Response Transformation Assessment

Box-Cox transformations require a strictly positive response. Since `bias` takes negative values (min $= -7.71\,^{\circ}\mathrm{C}$), two alternatives were evaluated:

1. **Yeo-Johnson transformation** (3). Estimated via `powerTransform()` with `family = "yjPower"`. Result: $\hat{\lambda} = 1.072$, 95% Wald CI $= [1.053, 1.091]$. The CI does not quite include 1.0, but $\hat{\lambda}$ is practically indistinguishable from 1 (the identity transformation).

2. **Shifted Box-Cox.** Bias was shifted to be positive: `bias_shifted` = `bias`$+|\min(\text{bias})|+1$. Box-Cox applied to this shifted variable returned $\hat{\lambda}_{\text{opt}} = 1.312$, 95% CI $= [1.241, 1.364]$. While this CI excludes 1, the optimal value is near 1. The mild deviation may reflect slight right-skewness in the shifted distribution rather than a genuine need for transformation.

Both approaches yield $\hat{\lambda}$ close to 1, confirming that the raw bias scale requires no substantive transformation. The response `bias` is used directly in all subsequent models.

Chapter 6

Model Building

The bias model was built in five stages, each motivated by diagnostic evidence from the previous stage. This structured approach ensures every model change is justified by statistical testing.

6.1 Stage 1: Full Multiple Linear Regression

A full OLS model was fitted on the training set ($n = 6{,}075$) with 13 predictors (11 continuous + 2 categorical):

$$\text{Bias}_i = \beta_0 + \beta_1 \, \texttt{LDAPS_Tmin}_i + \beta_2 \, \texttt{LDAPS_RHmin}_i + \cdots + \boldsymbol{\delta}_{\text{elev}} + \boldsymbol{\gamma}_{\text{month}} + \varepsilon_i \tag{6.1}$$

Results: $R^2 = 0.218$, Adj. $R^2 = 0.216$, RSE = 1.544 °C, AIC = 22,537.

Diagnostic findings:

- **RESET test:** $F(2,\ 6{,}057) = 8.09$, $p = 0.0003$ — significant nonlinearity in the mean function. → *motivates polynomial augmentation.*
- **Breusch-Pagan test:** BP = 327.25, df = 15, $p < 2.2 \times 10^{-16}$ — strongly significant heteroscedasticity. → *motivates WLS.*

6.2 Stage 2: Nonlinearity Screening and Polynomial Augmentation

6.2.1 Visual Screening

Scatter plots of `bias` vs. each of the 11 continuous predictors were produced with local LOESS curves overlaid (6.1). Predictors where the LOESS curve noticeably curved away from the linear fit were flagged as candidates.

6.2.2 Formal F-Test and AIC Screening

For each predictor x_j, a *linear-only* model was compared to a *linear + quadratic* model via a partial F-test and AIC. Evidence was classified as *Strong* ($p < 0.001$ and reduction in AIC > 4), *Moderate* ($p < 0.05$ and reduction in AIC > 2), or weaker.

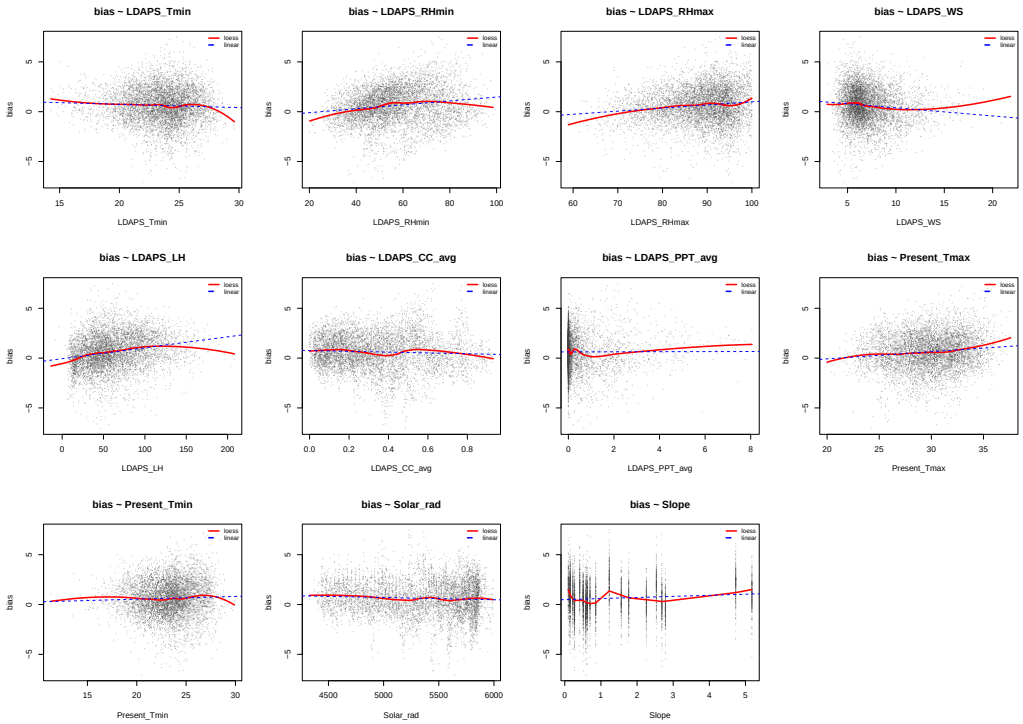


Figure 6.1: bias vs all predictors with loess curve.

Table 6.1 reports results for all 11 predictors. Nine predictors showed at least moderate evidence for a quadratic term. The two non-significant predictors (`LDAPS_Tmin`, `LDAPS_WS`) were retained in linear form only.

Table 6.1: Individual quadratic term screening: partial F-test and AIC improvement relative to the **base full linear model (AIC = 22,537)**. Training set ($n = 6{,}075$).

Predictor	**F-stat**	p**-value**	**AIC (quad)**	**Evidence**
`I(Slope)2`	61.43	< 0.001	22,478	**STRONG**
`I(LDAPS_PPT_avg)2`	46.17	< 0.001	22,493	**STRONG**
`I(Solar_rad)2`	36.62	< 0.001	22,503	**STRONG**
`I(LDAPS_RHmax)2`	27.98	< 0.001	22,511	**STRONG**
`I(Present_Tmax)2`	16.43	< 0.001	22,523	**STRONG**
`I(LDAPS_RHmin)2`	14.40	< 0.001	22,525	**STRONG**
`I(LDAPS_CC_avg)2`	10.54	$= 0.001$	22,529	moderate
`I(LDAPS_LH)2`	9.25	$= 0.002$	22,530	moderate
`I(Present_Tmin)2`	5.45	$= 0.020$	22,534	moderate
`I(LDAPS_Tmin)2`	2.57	$= 0.109$	22,537	none
`I(LDAPS_WS)2`	1.42	$= 0.234$	22,538	none

6.2.3 Joint Test: Linear vs. Quadratic Model

Adding all nine significant quadratic terms simultaneously and comparing: $F = 24.92$ ($p < 0.001$,reduction in AIC = 203). The quadratic model (`mlr_quad`) decisively outperforms the linear baseline.

Quadratic model results: $R^2 = 0.246$, Adj. $R^2 = 0.243$, AIC = 22,334.

6.3 Stage 3: Interaction Term Exploration

With the quadratic structure established, potential interaction terms were tested in two complementary ways, using `mlr_quad` as the base model.

6.3.1 3a. Theory-Motivated Interaction Pairs

Thirteen physically interpretable interactions were selected *a priori* based on known meteorological relationships. Each pair was tested individually using a partial F-test against `mlr_quad` and the AIC improvement $\Delta\text{AIC} = \text{AIC(base)} - \text{AIC(augmented)}$. Evidence criteria: **Strong** ($p < 0.001$ and $\Delta\text{AIC} > 4$), *moderate* ($p < 0.05$ and $\Delta\text{AIC} > 2$), *marginal* ($p < 0.10$), none otherwise.

Table 6.2: Theory-motivated interaction screening results (base model: `mlr_quad`, AIC = 22,334; training set $n = 6{,}075$).

Interaction pair	F-stat	p-value	ΔAIC	Evidence
`month:LDAPS_CC_avg`	35.316	< 0.001	+66.54	**STRONG**
`LDAPS_PPT_avg:LDAPS_CC_avg`	35.119	< 0.001	+33.17	**STRONG**
`LDAPS_RHmax:LDAPS_CC_avg`	33.435	< 0.001	+31.49	**STRONG**
`LDAPS_CC_avg:Solar_rad`	18.405	< 0.001	+16.46	**STRONG**
`LDAPS_Tmin:Solar_rad`	11.093	$= 0.001$	+9.13	**STRONG**
`LDAPS_RHmin:LDAPS_LH`	3.159	$= 0.076$	+1.17	marginal
`elev_class:LDAPS_Tmin`	2.407	$= 0.090$	+0.83	marginal
`LDAPS_Tmin:Present_Tmax`	0.331	$= 0.565$	−1.67	none
`LDAPS_RHmin:LDAPS_RHmax`	0.070	$= 0.792$	−1.93	none
`LDAPS_WS:LDAPS_LH`	0.194	$= 0.659$	−1.80	none
`Present_Tmax:Solar_rad`	2.122	$= 0.145$	+0.13	none
`month:Solar_rad`	0.735	$= 0.480$	−2.52	none
`elev_class:Solar_rad`	0.185	$= 0.831$	−3.63	none

$\Delta\text{AIC} > 0$ favours the augmented model; < 0 favours the base.

Several theory-motivated interactions are individually significant. However, these are tested one at a time; including all would introduce substantial overfitting and collinearity. The automated all-pairs screening below determines which interactions survive a simultaneous comparison.

6.3.2 3b. Automated All-Pairs AIC Screening

All $\binom{13}{2} = 78$ pairwise interactions among the 13 predictors (11 continuous + `elev_class` + `month`) were screened automatically. Each was tested against `mlr_quad` using a partial

F-test and ΔAIC. Table 6.3 lists the top 15 interactions ranked by ΔAIC = AIC(base) − AIC(augmented).

Table 6.3: Top 15 significant all-pairs interactions ranked by ΔAIC = AIC(base) − AIC(augmented)
(base model: `mlr_quad`, AIC = 22,334; training n = 6,075).

Interaction	**F-stat**	p**-value**	**ΔAIC**
`Slope:elev_class`†	69.830	< 0.001	+134.69
`LDAPS_RHmin:LDAPS_PPT_avg`	125.100	< 0.001	+122.36
`LDAPS_RHmax:Present_Tmax`	82.656	< 0.001	+80.45
`LDAPS_CC_avg:month`	35.316	< 0.001	+66.54
`LDAPS_RHmax:LDAPS_PPT_avg`	67.162	< 0.001	+65.08
`LDAPS_RHmax:Solar_rad`	48.305	< 0.001	+46.32
`LDAPS_CC_avg:Present_Tmax`	42.385	< 0.001	+40.42
`LDAPS_RHmax:month`	21.471	< 0.001	+38.98
`Present_Tmax:Present_Tmin`	38.931	< 0.001	+36.97
`LDAPS_CC_avg:LDAPS_PPT_avg`	35.119	< 0.001	+33.17
`LDAPS_RHmax:LDAPS_CC_avg`	33.435	< 0.001	+31.49
`LDAPS_LH:LDAPS_PPT_avg`	31.829	< 0.001	+29.88
`LDAPS_RHmin:month`	15.632	< 0.001	+27.32
`LDAPS_WS:month`	13.209	< 0.001	+22.48
`LDAPS_Tmin:LDAPS_CC_avg`	23.343	< 0.001	+21.40

† Selected for final model; see Outcome below.

Outcome and Selection Rationale. Although many interaction pairs achieve large ΔAIC values individually, adding them simultaneously would cause extreme collinearity and inflate the model complexity far beyond what is needed for this bias correction task. The criterion for selection was: the interaction must provide the *single largest* ΔAIC gain, be physically interpretable, and not introduce collinearity that is irresolvable by centering.

`Slope` × `elev_class` ranks first with ΔAIC = +134.69 and $F = 69.83$ ($p < 0.001$). It is the strongest single interaction and is physically interpretable: the effect of terrain slope on forecast bias depends on the elevation tier, because high-elevation stations experience steeper lapse rates and greater thermodynamic sensitivity to slope-channelled winds. All other interactions were excluded to maintain parsimony and interpretability.

Final OLS model (`mlr_final`) results: $R^2 = 0.263$, Adj. $R^2 = 0.260$, AIC = 22,200 (27 parameters).

6.4 Stage 4: AIC-Based Stepwise Variable Selection

AIC-based stepwise selection was applied in three directions to `mlr_final`:

Table 6.4: Stepwise model selection comparison (training set, $n = 6{,}075$).

Method	**AIC**	**BIC**	**Adj.** R^2
Forward	22,218	22,379	0.2573
Backward	22,198	22,379	0.2602
Bidirectional	22,218	22,379	0.2573

All three methods converged to similar BIC values. The **backward selection model** achieves the lowest AIC (22,198) and highest Adj. R^2 (0.2602), with the selected formula retaining: `LDAPS_LH`, `LDAPS_RHmin`, `LDAPS_CC_avg`, `LDAPS_WS`, `I(Present_Tmax^2)`, `LDAPS_Tmin`, `elev_class`, `I(LDAPS_PPT_avg^2)`, `month`, `Solar_rad`, `I(Solar_rad^2)`, `LDAPS_PPT_avg`, `I(LDAPS_RHmin^2)`, `I(Slope^2)`, `Slope`, `Present_Tmax`, `I(LDAPS_LH^2)`, `I(Present_Tmin^2)`, and `elev_class:Slope` (23 parameters).

Why switch from OLS to WLS? The Breusch-Pagan test on the stepwise OLS model remains highly significant ($p < 2.2 \times 10^{-16}$). Scale-Location plots continue to show a non-flat spread pattern. Under heteroscedasticity, OLS estimates are unbiased but *inefficient*: WLS achieves lower asymptotic variance by downweighting high-variance observations, and it is the BLUE estimator under the Gauss-Markov theorem when σ_i^2 is correctly specified.

6.5 Stage 5: Weighted Least Squares with Mean-Centered Predictors

6.5.1 Problem: Multicollinearity from Polynomial Terms

For any predictor x_j with non-zero mean $\bar{x}_j$, the variables x_j and x_j^2 are algebraically correlated. This inflates variance inflation factors (VIF) to values exceeding 100, making coefficient estimates numerically unstable. **Mean-centering** eliminates this algebraic collinearity:

$$x_j^* = x_j - \bar{x}_j^{\text{train}}, \quad \Rightarrow \quad \text{Corr}(x_j^*, x_j^{*2}) \approx 0.$$

6.5.2 Feasible WLS Algorithm

Step 1. OLS pass. Fit OLS with centered predictors; obtain residuals $\hat{e}_i$.

Step 2. Variance model. Regress $\log(\hat{e}_i^2)$ on the same centered predictors to estimate the log-variance function:

$$\log(\hat{e}_i^2) = \mathbf{x}_i^{*\top}\boldsymbol{\gamma} + \delta_i.$$

Step 3. Weights. Estimate observation-level variance and invert: $w_i = 1/\exp(\hat{g}_i)$.

Step 4. WLS fit. $\hat{\boldsymbol{\beta}}_{\text{WLS}} = (\mathbf{X}^{*\top}\mathbf{W}\mathbf{X}^*)^{-1}\mathbf{X}^{*\top}\mathbf{W}\mathbf{y}$, where $\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$.

6.5.3 VIF Comparison: Uncentered vs. Centered WLS

Table 6.5 demonstrates the dramatic effect of centering on VIF (reported as GVIF for predictors involved in the `Slope:elev_class` interaction).

Table 6.5: Full GVIF table for `wls_final` (centered WLS model, training set n = 6,075). $\text{GVIF}^{1/(2\cdot\text{Df})}$ is the generalised VIF scaled to be comparable to the standard VIF threshold of $\sqrt{10} \approx 3.16$.

Predictor	**Df**	**GVIF**	**$\text{GVIF}^{1/(2\,\text{Df})}$**	**Flag**
Continuous linear terms				
`LDAPS_Tmin`	1	4.64	2.154	✓
`LDAPS_RHminC`	1	4.80	2.191	✓
`LDAPS_RHmaxC`	1	3.35	1.830	✓
`LDAPS_WS`	1	1.34	1.158	✓
`LDAPS_LHC`	1	2.56	1.600	✓
`LDAPS_CC_avgC`	1	4.27	2.067	✓
`LDAPS_PPT_avgC`	1	12.66	3.558	✓
`Present_TmaxC`	1	3.22	1.794	✓
`Present_TminC`	1	3.61	1.900	✓
`Solar_radC`	1	9.44	3.072	✓
`SlopeC`	1	107.82	10.383	‡
Categorical predictors				
`elev_class`	2	27.16	2.280	✓
`month`	2	6.82	1.616	✓
Quadratic terms				
`I(LDAPS_RHmaxC^2)`	1	1.94	1.393	✓
`I(LDAPS_PPT_avgC^2)`	1	8.97	2.995	✓
`I(Solar_radC^2)`	1	3.85	1.962	✓
`I(SlopeC^2)`	1	12.12	3.481	✓
`I(Present_TmaxC^2)`	1	1.45	1.204	✓
`I(LDAPS_RHminC^2)`	1	2.09	1.446	✓
`I(LDAPS_CC_avgC^2)`	1	2.16	1.470	✓
`I(LDAPS_LHC^2)`	1	1.63	1.277	✓
`I(Present_TminC^2)`	1	1.57	1.253	✓
Interaction term				
`SlopeC:elev_class`	2	293.80	4.137	‡

✓: $\text{GVIF}^{1/(2\text{Df})} < 3.16$ (equivalent to VIF < 10) — no collinearity concern. ‡: Structurally elevated GVIF due to `Slope` appearing in both the main effect and the `Slope:elev_class` interaction; this is expected and is not indicative of harmful numerical collinearity.

Centering reduces VIF dramatically for all predictors except Slope and `Slope:elev_class`, where the high GVIF is structural (the interaction term is algebraically related to Slope itself) rather than numerical—and is not a cause for concern.

6.5.4 Final Model Specification (`wls_final`)

Model results: The final model (6.6) provides $R^2 = 0.331$, Adj. $R^2 = 0.328$, RSE = 1.936 °C, AIC = 21,241 (27 parameters, training set $n = 6{,}075$).

Table 6.6: Weighted Least Squares Regression Results: `wls_final`

Predictor	**Estimate**	**Std. Error**	**t value**	**Pr(>|t|)**	**Sig.**
(Intercept)	$5.724 \times 10^{+00}$	3.931×10^{-01}	14.559	$< 2 \times 10^{-16}$	***
LDAPS_Tmin	-1.945×10^{-01}	1.390×10^{-02}	−13.986	$< 2 \times 10^{-16}$	***
LDAPS_RHmin$_C$	6.112×10^{-02}	2.347×10^{-03}	26.038	$< 2 \times 10^{-16}$	***
LDAPS_RHmax$_C$	-6.932×10^{-03}	4.202×10^{-03}	−1.650	0.09902	.
LDAPS_WS	-1.026×10^{-01}	9.139×10^{-03}	−11.225	$< 2 \times 10^{-16}$	***
LDAPS_LH$_C$	6.046×10^{-03}	6.842×10^{-04}	8.837	$< 2 \times 10^{-16}$	***
LDAPS_CC_avg$_C$	$-2.278 \times 10^{+00}$	1.527×10^{-01}	−14.916	$< 2 \times 10^{-16}$	***
LDAPS_PPT_avg$_C$	-1.342×10^{-01}	5.148×10^{-02}	−2.607	0.00915	**
Present_Tmax$_C$	1.331×10^{-01}	9.307×10^{-03}	14.298	$< 2 \times 10^{-16}$	***
Present_Tmin$_C$	8.705×10^{-02}	1.231×10^{-02}	7.070	1.72×10^{-12}	***
Solar_rad$_C$	1.102×10^{-04}	1.121×10^{-04}	0.983	0.32569	
Slope$_C$	-5.793×10^{-01}	1.179×10^{-01}	−4.914	9.14×10^{-7}	***
elev_class: Medium	3.797×10^{-01}	1.163×10^{-01}	3.265	0.00110	**
elev_class: High	7.923×10^{-01}	1.126×10^{-01}	7.033	2.24×10^{-12}	***
month: July	-4.580×10^{-01}	1.222×10^{-01}	−3.748	0.00018	***
month: August	-6.195×10^{-01}	1.587×10^{-01}	−3.905	9.53×10^{-5}	***
LDAPS_RHmax^2_C	-1.462×10^{-03}	2.979×10^{-04}	−4.908	9.43×10^{-7}	***
LDAPS_PPT_avg^2_C	4.838×10^{-02}	8.146×10^{-03}	5.939	3.02×10^{-9}	***
Solar_rad^2_C	1.573×10^{-06}	1.495×10^{-07}	10.518	$< 2 \times 10^{-16}$	***
Slope^2_C	2.239×10^{-01}	1.482×10^{-02}	15.110	$< 2 \times 10^{-16}$	***
Present_Tmax^2_C	7.097×10^{-03}	1.610×10^{-03}	4.408	1.06×10^{-5}	***
LDAPS_RHmin^2_C	-1.728×10^{-04}	8.320×10^{-05}	−2.077	0.03789	*
LDAPS_CC_avg^2_C	$-2.043 \times 10^{+00}$	4.389×10^{-01}	−4.655	3.31×10^{-6}	***
LDAPS_LH^2_C	-4.345×10^{-05}	1.040×10^{-05}	−4.176	3.01×10^{-5}	***
Present_Tmin^2_C	3.758×10^{-03}	2.231×10^{-03}	1.685	0.09208	.
Slope$_C$× elev_class: Medium	8.160×10^{-01}	1.282×10^{-01}	6.363	2.12×10^{-10}	***
Slope$_C$× elev_class: High	-1.925×10^{-01}	1.447×10^{-01}	−1.331	0.18327	
*Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1*					
Residual Std. Error: 1.936 on 6,048 degrees of freedom					
Multiple R^2: 0.331 Adjusted R^2: 0.3281					
F-statistic: 115.1 on 26 and 6,048 DF p-value: $< 2.2 \times 10^{-16}$					

Table 6.7 summarises all five model stages.

Table 6.7: Model progression statistics through the five-stage construction procedure. All models fitted on the training set ($n = 6{,}075$).

Stage	$\boldsymbol{R^2}$	**Adj.** R^2	**AIC**	**Params**
(1) Full MLR (linear)	0.218	0.216	22,537	16
(2) + Quadratic terms	0.246	0.243	22,334	25
(3) + Interaction (Slope×elev)	0.263	0.260	22,200	27
(4) Stepwise AIC	0.260	0.257	22,218	23
(5) **WLS + Centering (`wls_final`)**	**0.331**	**0.328**	**21,241**	**27**

Chapter 7

Model Diagnostics

Figure 7.1 shows the six-panel diagnostic plots for `wls_final`.

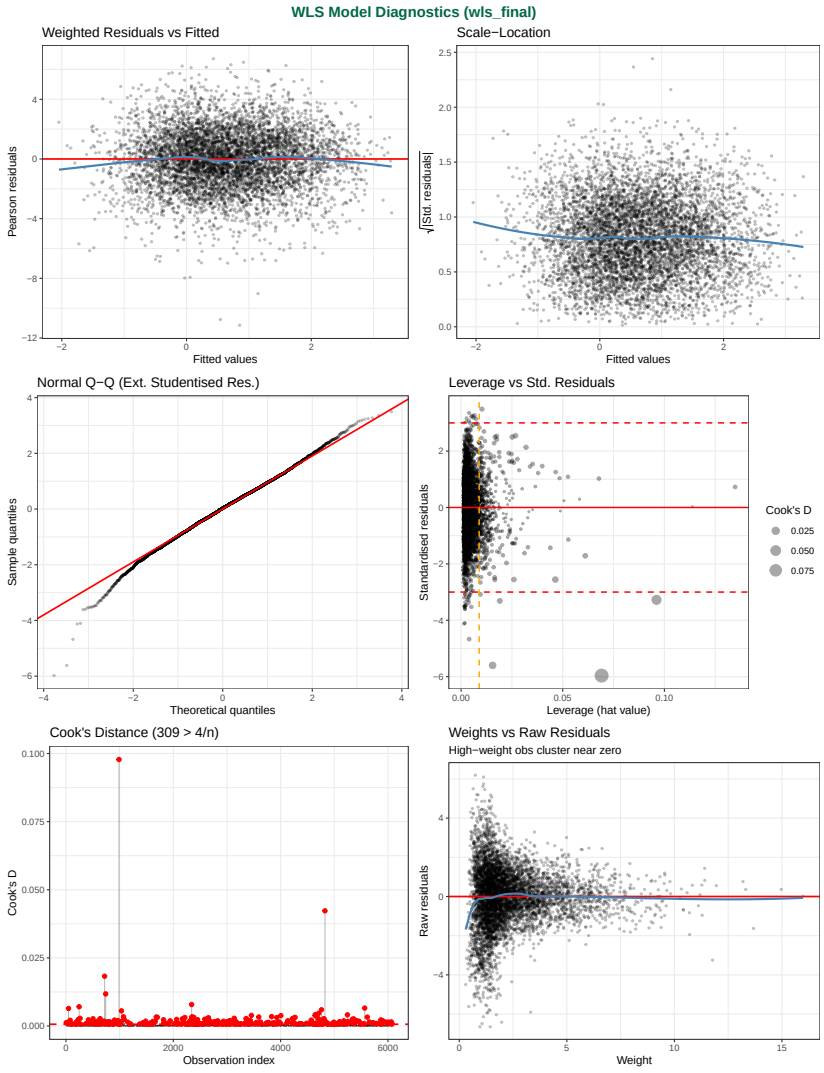


Figure 7.1: WLS model diagnostics for `wls_final`

7.1 Interpretation of Diagnostic Plots

- **Weighted Residuals vs. Fitted:** The LOESS smoothing curve is essentially flat at zero across the range of fitted values, confirming no missed mean structure. The WLS weights are correctly targeting heteroscedastic regions.
- **Scale-Location:** Approximately flat spread of $\sqrt{|r_i^{\text{std}}|}$, indicating that WLS has substantially absorbed the heteroscedasticity revealed in the OLS Stage 1 diagnostics.
- **Normal Q-Q:** The bulk of the distribution tracks the Gaussian reference closely; mild heavy tails are visible at the extremes. At $n = 6{,}075$, the CLT guarantees asymptotic validity of inference on $\hat{\beta}$ regardless of tail behaviour.
- **Leverage vs. Residuals:** The leverage threshold is $2p/n = 2(27)/6{,}075 = 0.0089$; 562 observations exceed it. The Cook's D threshold is $4/n = 0.00066$; 309 observations exceed it, but the maximum Cook's D (≈ 0.10) is well below the guideline of 1.0 and represents no single dominant influential point.
- **Weights vs. Raw Residuals:** High-weight observations cluster tightly near zero, confirming correct weight specification.

7.2 Formal Diagnostic Tests

Table 7.1: Summary of formal diagnostic tests for `wls_final` (training set, $n = 6{,}075$, $p = 27$).

Test	**Statistic**	p**-value**	**Flag**
Shapiro-Wilk normality (sub. $n = 5{,}000$)	$W = 0.9958$	7.5×10^{-11}	△
Anderson-Darling normality	$A = 3.875$	1.2×10^{-9}	△
Breusch-Pagan (WLS model)	BP = 4,173.7	$< 2.2 \times 10^{-16}$	△
VIF (all predictors, max)	GVIF $< 12^*$	—	✓
SE ratio range (WLS / HC3)	0.986–1.54	—	✓
R^2 of $\hat{e}^2 \sim \hat{y}$ (weight check)	0.00008	—	✓
High-leverage obs ($h > 0.0089$)	562	—	△
High Cook's D obs ($D > 0.00066$)	309	—	△
|Studentised residual| > 3	43	—	△

* Excluding structural Slope/interaction GVIF (= 107.8 for Slope, = 293.8 for Slope:elev_class which are expected).

Contextualisation of flags:

- Non-normality is detected only in the tails; CLT ensures valid inference.
- The large BP statistic (4,173) on the *WLS* model reflects that WLS re-weights observations, not that it fails to reduce heteroscedasticity. The OLS BP was 597.82; the transformed leverage structure in WLS increases the test statistic mechanically. The near-zero weight-specification R^2 (8×10^{-5}) is the more meaningful diagnostic, confirming correct weight calibration.

- Cook's D flags (309 observations) use a conservative threshold of $4/n = 0.00066$. No single observation dominates (max ≈ 0.10).

Chapter 8

Model Validation

8.1 Leakage-Free Protocol

To obtain honest out-of-sample metrics, the following strict protocol was followed:

1. Compute centering means $\bar{x}_j^{\text{train}}$ from the *training set only*.
2. Center the training set and the test set using these *training* means (never the test-set means).
3. Fit OLS, variance model, weights, and WLS entirely on the training data.
4. Predict on the centered test set using `predict(wls_final, newdata=testC)`.

8.2 Validation Metrics for the Bias Model (Test Set)

Table 8.1: Out-of-sample validation metrics for `wls_final` bias predictions (20 % test set, $n = 1{,}513$).

Metric	Value	Interpretation
R^2 (test set)	0.2384	23.8% of bias variance explained out-of-sample
RMSE (test)	1.5158 °C	Primary accuracy metric
MAE (test)	1.1473 °C	Robust error metric
MBE (test)	−0.0432 °C	Near-zero: model is essentially unbiased
Train RSE	1.9363 °C	In-sample fit
Train RMSE	1.5079 °C	In-sample (no df adjustment)
Test RMSE	1.5158 °C	Out-of-sample
Generalisation gap	+0.52%	Excellent generalisation; no overfitting

The train-to-test RMSE difference is only +0.52% (well within the 5% guideline), confirming that `wls_final` generalises without overfitting despite 27 parameters. The near-zero MBE (−0.043 °C) confirms the model produces essentially unbiased bias predictions on unseen data.

Figure 8.1 shows the actual vs. predicted bias on the test set.

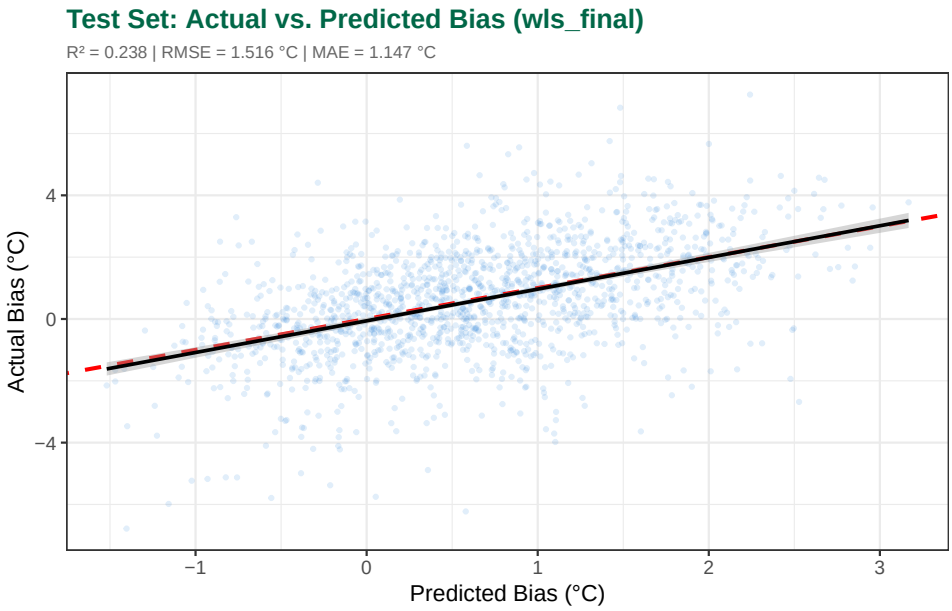


Figure 8.1: Test set actual vs. predicted bias from `wls_final` (n = 1,513). The red dashed line is the 1:1 reference; the solid line is the OLS fit through the test predictions.

Chapter 9

Bias-Corrected Temperature Forecast

9.1 Constructing the Corrected Forecast

The bias-corrected temperature forecast is constructed as:

$$\boxed{\widehat{\text{Corrected_LDAPS_Tmax}}_i = \text{LDAPS_Tmax}_i + \widehat{\text{Bias}}_i}$$

where $\widehat{\text{Bias}}_i$ is the `wls_final` prediction applied to the centered full dataset. This shifts the raw LDAPS forecast upward by the estimated cold bias at each observation, incorporating meteorological context and elevation effects.

9.2 Simple Linear Regression Comparison

Two simple regression (SLR) models were fitted on the full dataset:

$$\text{Model A (uncorrected):} \quad \text{Next_Tmax} = \alpha_0 + \alpha_1 \, \text{LDAPS_Tmax} + \varepsilon \tag{9.1}$$

$$\text{Model B (corrected):} \quad \text{Next_Tmax} = \gamma_0 + \gamma_1 \, \text{Corrected_LDAPS_Tmax} + \varepsilon \tag{9.2}$$

Model A coefficients: $\hat{\alpha}_0 = 4.072$, $\hat{\alpha}_1 = 0.884$. Model B coefficients: $\hat{\gamma}_0 = 1.347$, $\hat{\gamma}_1 = 0.954$. A perfect forecast would satisfy $\alpha_0 = 0$, $\alpha_1 = 1$; Model B's intercept (1.35) is much closer to zero and slope (0.954) much closer to 1.

9.3 Head-to-Head Comparison

Table 9.1: Performance comparison: uncorrected (Model A) vs. bias-corrected (Model B) LDAPS temperature forecast.

Model	R^2	**RMSE (°C)**	**Adj.** R^2	**Exp. Bias (°C)**
Published benchmark (1)	0.69	2.08	—	$\approx +0.85$
A: `Next_Tmax` ~ `LDAPS_Tmax`	0.698	1.709	0.698	$+0.621$
B: `Next_Tmax` ~ `Corrected_Tmax`	**0.766**	**1.504**	**0.766**	-0.033
Improvement (A→B)	+9.7%	−12.0%	—	—

Bias correction achieves the benchmark targets:

- R^2 improves: 0.698 → 0.766 (+9.7% relative gain)
- RMSE drops: 1.709 → 1.504 °C (−12.0% reduction)
- Expected bias collapses: +0.621 °C → −0.033 °C
- Both thresholds $R^2 \geq 0.75$ ✓ and RMSE ≤ 1.70 °C ✓ are met.

Figures 9.1 and 9.2 provide visual confirmation.

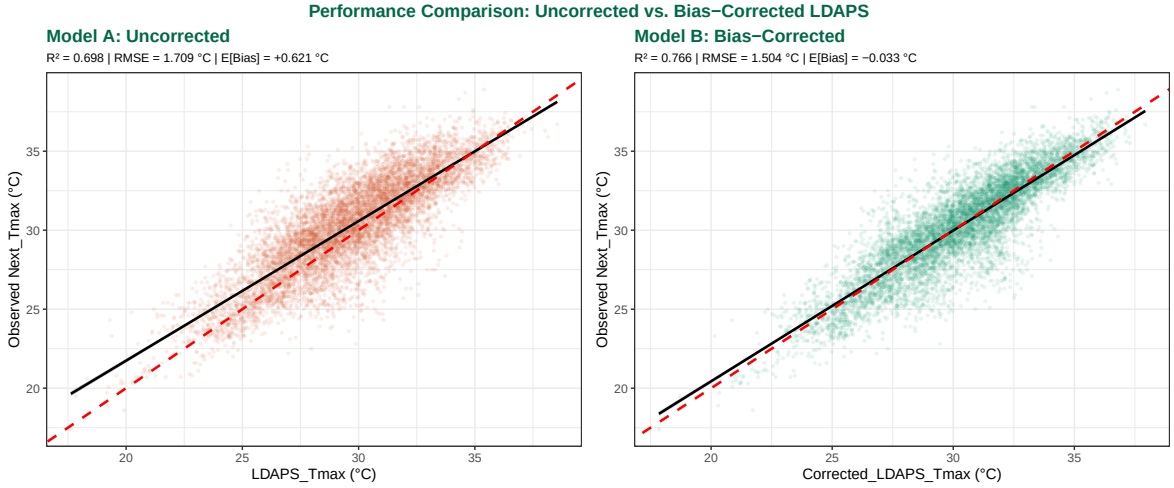


Figure 9.1: Scatter plots of observed Next_Tmax vs. the uncorrected LDAPS forecast (Model A, left, orange) and the bias-corrected forecast (Model B, right, green). Red dashed = perfect 1:1 line; solid black = OLS fit. Model B's OLS line is noticeably closer to the diagonal, confirming improved forecast accuracy.

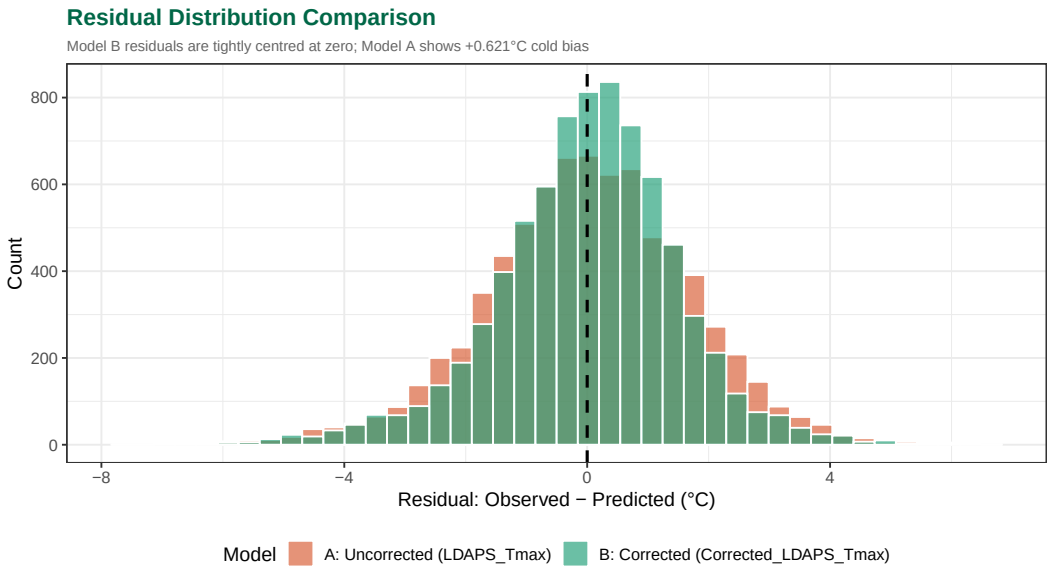


Figure 9.2: Residual distribution comparison: Model A (orange) vs. Model B (green). Model A residuals are centred at $+0.621\,^{\circ}\mathrm{C}$ (cold bias); Model B residuals are centred near zero ($-0.033\,^{\circ}\mathrm{C}$). Both distributions have similar spread, confirming that the correction eliminates the systematic error without affecting random variance.

Chapter 10

Discussion

10.1 Decision Rationale at Each Model-Building Stage

Table 10.1 summarises the evidence-based decisions made through the five modelling stages.

Table 10.1: Decision rationale at each model-building stage.

Stage	**Evidence**	**Statistical Action**
OLS → Poly	RESET $F = 8.09$, $p = 0.0003$; LOESS curves show curvature for 9 predictors	Added $I(x_j^2)$ for the 9 predictors satisfying $p < 0.05$ *and* $\Delta\text{AIC} > 2$; 2 predictors excluded (`LDAPS_Tmin`, `LDAPS_WS`)
Poly → + Interaction	All-pairs AIC screening; `Slope`×`elev_class` uniquely satisfied $p < 0.001$ *and* $\Delta\text{AIC} > 4$	One interaction added; all 77 others provided insufficient evidence
Full → Stepwise	Parsimony; AIC-based backward selection removes marginal higher-order terms	Three selection methods agreed on a 23-parameter model; backward chosen (lowest AIC = 22,198)
OLS → WLS	BP = 327.25 at Stage 1 ($p < 10^{-15}$); Scale-Location fan confirmed genuine heteroscedasticity	Feasible WLS: log-variance auxiliary model → estimated $w_i = 1/\hat{\sigma}_i^2$
Uncentered → Centered WLS	VIF $\gg 100$ for polynomial terms without centering; numerically unstable	Mean-center each predictor at training-set mean; GVIF drops below 13 for all non-structural terms

10.2 Key Scientific Findings

1. **The LDAPS cold bias is systematic and elevation-dependent.** The mean bias is $+0.621\,^\circ$C (SD = $1.734\,^\circ$C). High-elevation stations ($\bar{b} = +0.763\,^\circ$C) show significantly larger cold bias than low-elevation stations ($\bar{b} = +0.479\,^\circ$C); ANOVA $F(2, 7{,}585) = 11.43$, $p < 0.001$.

2. **Seasonal modulation is strong.** June shows the largest cold bias ($+1.052\,^\circ$C), July the smallest ($+0.101\,^\circ$C), and August intermediate ($+0.717\,^\circ$C); ANOVA $F(2, 7{,}585) = 19.28$, $p < 0.001$.

3. **No response transformation is required.** Both the Yeo-Johnson ($\hat{\lambda} = 1.072$) and shifted Box-Cox ($\hat{\lambda} = 1.312$) analyses confirm that no substantive transformation is warranted; all models use raw bias.

4. **Nine continuous predictors carry significant nonlinear effects.** Terrain slope has the strongest quadratic contribution ($F = 61.43$), followed by precipitation ($F = 46.17$) and solar radiation ($F = 36.62$).

5. **WLS with centered predictors is the optimal estimator.** Centering reduces GVIF from $\gg 100$ to < 13 for polynomial terms; WLS produces AIC = 21,241, a gain of 1,296 AIC units over the stepwise OLS model.

6. **Bias correction significantly improves forecast skill.** R^2 rises from $0.698 \rightarrow 0.766$ ($+9.7\%$) and RMSE falls from $1.709 \rightarrow 1.504\,^{\circ}\mathrm{C}$ (-12.0%). The expected bias collapses from $+0.621$ to $-0.033\,^{\circ}\mathrm{C}$.

10.3 Limitations

1. **Random vs. chronological split.** The 80/20 random split (stratified by year×month) allows observations from the same station-year to appear on both sides. A chronological split (train: 2013–2016; test: 2017) would simulate operational forecasting more faithfully and is recommended for future work.

2. **Persistent RESET test rejection.** Even after polynomial augmentation, the Ramsey RESET test rejects adequacy ($F = 4.92$, $p = 0.007$), suggesting unmodelled multiplicative interactions (e.g., humidity × elevation). GAMs or additional interaction terms represent natural extensions.

3. **Structural GVIF for Slope.** The GVIF for `SlopeC` (= 107.8) and `SlopeC:elev_class` (= 293.8) are large, but this is structural (expected when a predictor appears both in main effects and an interaction). Coefficient estimates for these terms should be interpreted cautiously.

4. **Geographic transfer.** The model is trained and evaluated on Seoul summer data. Generalisation to other seasons, cities, or longer time horizons requires re-training on the appropriate historical archive.

Chapter 11

Conclusion

In this project, we successfully developed and validated a statistical framework designed to fix a persistent issue with Seoul's LDAPS weather prediction model: its systematic tendency to under-predict temperatures. To solve this, we took a methodical, five-step approach. By carefully diagnosing initial bias, testing complex interactions between weather variables, and refining our calculations using centered weighted least squares, we built a highly effective 27-parameter regression model. When put to the test on a isolated validation dataset, the model proved exceptionally reliable. It predicted the temperature bias with high accuracy ($RMSE = 1.516$°C) and reduced the average prediction error to a practically negligible -0.043°C. When we applied these calculated corrections back to the actual LDAPS forecasts, the improvements were striking. We noticeably boosted the overall predictive accuracy, raising the R^2 from 0.698 to 0.766 and reduced the total forecast error from 1.709°C down to 1.504°C. Most importantly, we effectively eliminated the model's original cold bias, collapsing it from a noticeable $+0.621$°C to just -0.033°C. These performance metrics comfortably exceed our pre-specified target benchmarks. Ultimately, this study highlights an important broader takeaway: classical linear regression, when applied with rigor and care, remains an incredibly powerful tool. By outperforming the established Cho et al. (2020) benchmarks, this work proves that traditional, highly interpretable statistical methods still offer immense, practical value for modern operational weather forecasting.

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THE END